

4(a)	Simple harmonic motion is defined as a periodic motion of a particle whose acceleration is directly proportional to its displacement from the equilibrium position and this acceleration is always directed towards that position.	[1] [1]
(b)(i)	$x_o = 22/2 = \mathbf{11 \text{ mm}}$	[1]-ans
(ii)	$\omega = 2\pi f = 2\pi(4.5) = \mathbf{28.3 \text{ rad s}^{-1}}$	[1]-ans
(iii)	$a = \omega^2 x_o = (28.32)^2 (0.011) = \mathbf{8.8 \text{ m s}^{-2}}$	[1]-ans
(iv)	$x = 11.0 - 8.0 = \mathbf{3.0 \text{ mm}}$	[1]-ans
MC	accept no d.p., not accept 2.d.p.	
(v)	$v = \omega \sqrt{(x_o^2 - x^2)} = 28.3 \sqrt{0.011^2 - 0.0030^2}$ $= \mathbf{0.30 \text{ m s}^{-1}}$	[1]-sub [1]-ans
5(a)(i)	Wavelength = 12 m	